

Year in Review

Global Internet Traffic, Technology Adoption,
Connectivity & Security Trends

Observed through Cloudflare's global network

17%

Traffic Growth

225

Disruptions

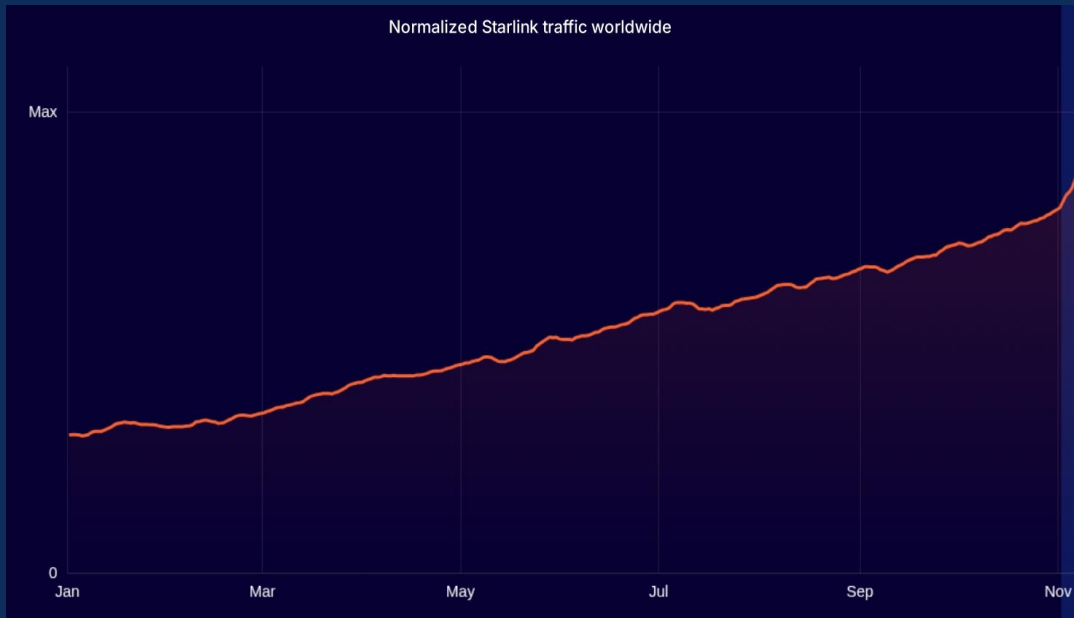
6.5%

Traffic Mitigated

13%

Post-Quantum TLS

17% Global Traffic Growth and Starlink's 3.3x Surge



Global Internet Traffic

+17% Year-over-Year

Continuing the upward trend driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation.

Starlink Satellite Traffic

3.3x

Growth in 2024

SpaceX Starlink brought satellite Internet connectivity to previously underserved locations worldwide. The trend line is scaled relative to peak observed request volume.

- ▶ Expanding to underserved and remote regions
- ▶ Growing consumer and enterprise adoption
- ▶ Complementing terrestrial infrastructure

Google Leads Top 10 Services as Generative AI Rises

Rank	Service	Category
#1	Google	Search, Ads, Cloud
#2	Facebook	Social Media
#3	Apple	Devices, Services
#4	TikTok	Short Video
#5	AWS	Cloud Infrastructure
#6	Microsoft	Software, Cloud
#7	Instagram	Social Media
#8	YouTube	Video
#9	Amazon	E-Commerce, Cloud
#10	WhatsApp	Messaging

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#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Generative AI Surge

Generative AI continued its meteoric rise throughout 2024, reshaping how users interact with Internet services. Cloudflare observed massive growth in AI-driven traffic and requests.

Metaverse technologies receded into the background as enterprise and consumer focus shifted to LLM-powered applications.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

13%

Post-Quantum TLS

- Up from ~2% in early 2024
- Chrome 124 enabled PQ by default
- April 2024

21%

HTTP/3 Traffic

- QUIC-based protocol
- Faster connections
- Encrypted by default

Additional Technology Metrics

89% Google Search market share

33% iOS traffic share
Peak iOS reached ~66% in some regions

28% IPv6 adoption
of dual-stack traffic globally

41% Mobile traffic
Share of total global traffic

Chrome
Most popular browser
Referral traffic leader

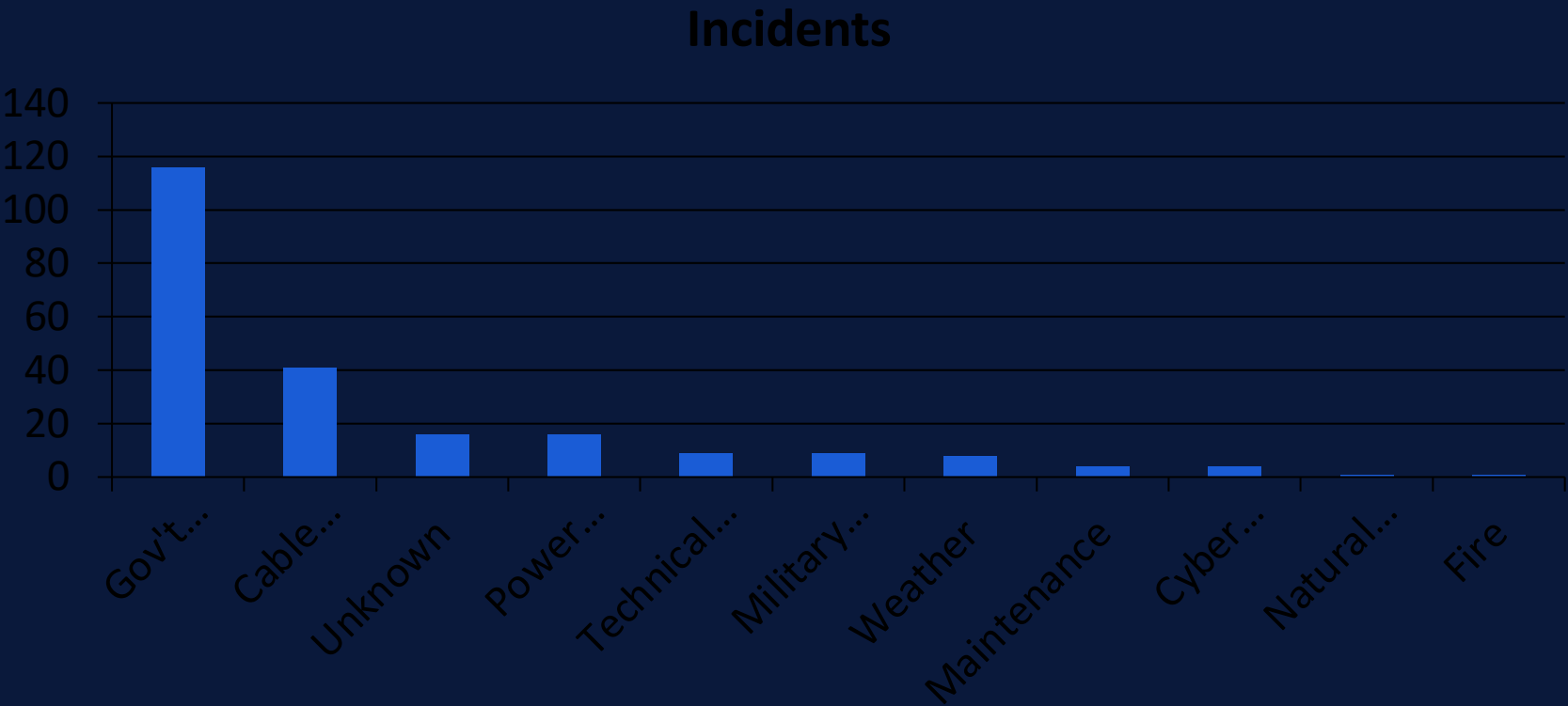
Post-quantum encrypted traffic worldwide



225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

Major Internet disruptions observed globally in 2024



Cause	Count	Cause	Count
Government Directed	116+	Weather	8
Cable Cut	41	Maintenance	4
Unknown	16	Cyber Attack	4
Power Outage	16	Natural Disaster	1
Technical Problem	9	Fire	1

6.5% of Traffic Mitigated: Persistent Threats & Bot Concentration

6.5%

Traffic Mitigated

Share of all Cloudflare traffic processed

68%

Bot Traffic from Top 10

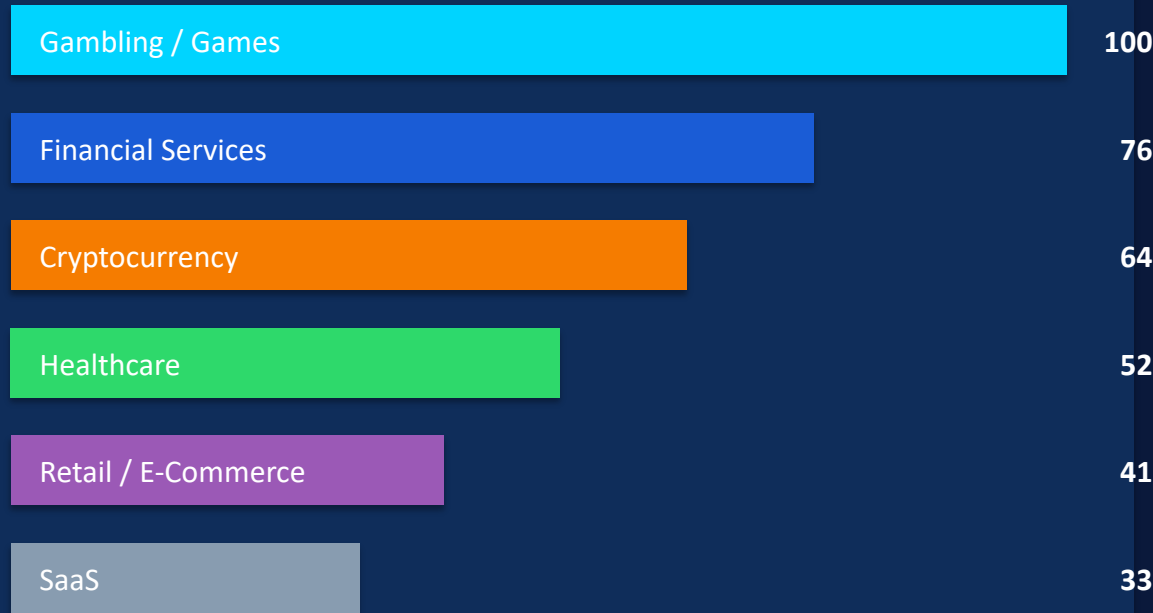
Global bot traffic originates from just 10 countries

4.3%

Malicious Emails

Share of processed emails containing threats

Most Attacked Sectors



Log4j Persistence (CVE-2021-44228)

Active exploitation continues 2+ years after disclosure. Widely known vulnerabilities remain a favoured attack vector despite available patches.

Email Security Threats

- ▶ 4.3% of emails processed were malicious
- ▶ 43% of malicious emails contained deceptive links
- ▶ .bar TLD originated the largest share of malicious email
- ▶ RPKI valid routes increased 6.4%, improving routing security

Five Takeaways Shaping the 2025 Internet Outlook

01

Traffic Growth Trajectory

Global Internet traffic grew 17% YoY with Starlink surging 3.3×, showing no sign of slowdown as connectivity becomes essential infrastructure for billions.

02

Post-Quantum Encryption Momentum

Post-quantum TLS jumped from ~2% to 13% after Chrome 124's default enablement. Security-conscious organisations are accelerating migration.

03

Connectivity Fragility

225 major disruptions observed; government-directed shutdowns remain the leading cause. Internet governance remains a critical risk factor.

04

Persistent Security Threats

6.5% of traffic was mitigated. Log4j still exploited 2+ years on. Gambling/Games most attacked sector. 68% of bot traffic from top 10 countries.

05

Evolving Developer Ecosystem

Go surpassed Node.js and Python as the top API client language (~12%). HTTP/3 at 21%. AI services are reshaping service rankings and traffic patterns.